12. Modular Arithmetic, Division Thm. Rationals and irrationals (3.7)

Last Time: For $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$, we say $a \equiv b \pmod{n}$ if $n \mid (a-b)$

Theorem 1: Let $n \in \mathbb{Z}^+$. The relation $_ \equiv _ \pmod{n}$ is:

1) Reflexive: For all $a \in \mathbb{Z}$, $a \equiv a \pmod{n}$
2) Symmetric: For all $a, b \in \mathbb{Z}$, if $a \equiv b \pmod{n}$ Then $b \equiv a \pmod{n}$
3) Transitive: For all $a, b, c \in \mathbb{Z}$, if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, Then $a \equiv c \pmod{n}$

Proof: HW 4 □

This Thm says That "Congruence mod n" behaves like equality in a weaker sense

Next: Why is it called Modular Arithmetic?

Theorem 2: Let $a, b, c, d \in \mathbb{Z}$ and $n \in \mathbb{N}$.
If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, Then:

1) $a + c \equiv b + d \pmod{n}$
2) $ac \equiv bd \pmod{n}$
3) $a^m \equiv b^m \pmod{n}$ for any $m \in \mathbb{N}$.

Example: 1) $29 \cdot 17 \equiv ?$ (mod 11)

Since $29 \equiv 7 \pmod{11}$ and $17 \equiv 6 \pmod{11}$ we have

$$29 \cdot 17 \equiv 7 \cdot 6 \equiv 42 \equiv 9 \pmod{11}$$

2) $(26)^5 \equiv ?$ (mod 24)

Since $26 \equiv 2 \pmod{24}$ $\quad 26^5 \equiv 2^5 \equiv 32 \equiv 8 \pmod{24}$

Proof of Thm 2: (1) and (3) HW!

(2) Fix $a, b, c, d \in \mathbb{Z}$, and $n \in \mathbb{Z}$. Suppose $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$.

By definition, $n \mid (a-b)$ and $n \mid (c-d)$. Hence, there exist $k_1, k_2 \in \mathbb{Z}$ such that

$$a - b = n k_1 \quad \text{and} \quad c - d = n k_2.$$

> Want: $ac \equiv bd \pmod{n}$
> ie: $n \mid (ac - bd)$

Multiplying $a - b = nk_1$ by c gives

$$ac - bc = n k_1 c$$

And $c - d = n k_2$ by b gives

$$bc - bd = n k_2 b$$

Adding these together, we get

$$ac - bd = n k_1 c + n k_2 b = n(k_1 c + k_2 b)$$

By closure of $\mathbb{Z}$, $k_1 c + k_2 b \in \mathbb{Z}$, so $n \mid (ac - bd)$, which means $ac \equiv bd \pmod{n}$

□

Division Theorem: For any $a \in \mathbb{Z}$ and $n \in \mathbb{N}$, there exist unique integers q and r such that

$$a = nq + r \quad \text{with} \quad 0 \leq r < n.$$

q is called the quotient and r is called the reminder

Note:

For $n = 2$, this in particular implies that any $a \in \mathbb{Z}$ is either odd ($r = 1$) or even ($r = 0$) but not both (r is unique)

Example: $a = 93$, $n = 6$. Then $q = 15$ and $r = 3$

$$93 = 15 \cdot 6 + 3$$

For $a = -16$, $n = 6$, we have $-16 = 6 \cdot (-3) + 2$ so $q = -3$, $r = 2$

Corollary: For any $a \in \mathbb{Z}$ and $n \in \mathbb{N}$, there is a unique integer r with $0 \leq r < n$ such that

$$a \equiv r \pmod{n}$$

Rationals / Irrationals

Definition 4: Let $x \in \mathbb{R}$. We say that x is **rational** if there exist $p, q \in \mathbb{Z}$ with $q \neq 0$ such that $x = pq^{-1}$. The set of all rationals is denoted by $\mathbb{Q}$.
We say that x is **irrational** if it is not rational.

Theorem 5: Let $n \in \mathbb{Z}$. Then n is even if and only if n^2 is even.

Proof: [$\Rightarrow$] Exercise

[$\Leftarrow$] Fix $n \in \mathbb{Z}$. Want to show: if n^2 is even then n is even $\quad P \to Q$
Let us show the contrapositive: if n is not even then n^2 is not even.
By thm 10.4, "not even" just means "odd".
So we want to show: if n is odd then n^2 is odd.
Assume n is odd, so $n = 2k+1$ for some $k \in \mathbb{Z}$. But then

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

where $2k^2 + 2k$ is an integer by closure, thus n^2 is odd. $\square$

Technique: (Contrapositive) Instead of proving $P \to Q$, prove $\neg Q \to \neg P$.

Theorem 6: If x is a real number such that $x^2 = 2$, then x is irrational.

Proof: Let $x \in \mathbb{R}$ such that $x^2 = 2$. Suppose for a contradiction that x is rational.

Thus $x = pq^{-1}$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$.

Also we can choose $p \in \mathbb{Z}^+$.

By well-ordering, there is a smallest such p.

Then $x = pq^{-1}$ implies $qx = p$ so $q^2x^2 = p^2$, i.e: $2q^2 = p^2$.

Therefore p^2 is even, so also p is even. Thus $p = 2k$ for some $k \in \mathbb{Z}$.

Now $p^2 = 2q^2$ gives $4k^2 = 2q^2$, so $2k^2 = q^2$, which says q^2 even.

Then also q is even. So we can reduce $x = \frac{p}{q}$ to get a smaller p,

contradiction. □